

**B.Tech. Degree I Semester Regular/Supplementary and
II Semester Supplementary Examination March 2022**

CS/EC/IT 19-200-0103B & CE/EE/ME/SE 19-200-0203A

ENGINEERING MECHANICS

(2019 Scheme)

Time: 3 Hours

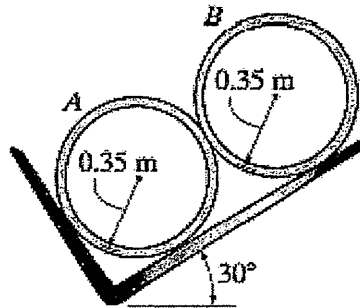
Maximum Marks: 60

PART A

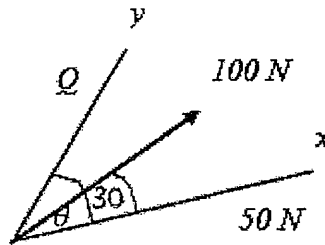
(Answer *ALL* questions)

(8 × 3 = 24)

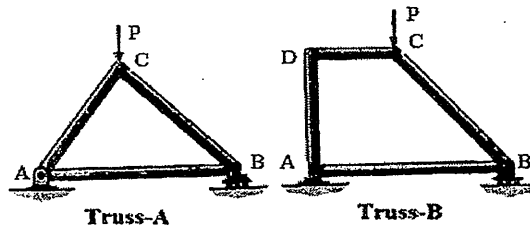
- I. (a) Two smooth pipes, each having a mass of 300 kg, are supported by the forked tines of a tractor as shown in figure. Draw the free body diagram of the two pipes together and then each pipe separately.



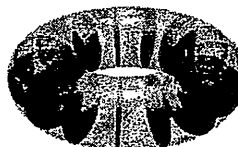
- (b) If one component of a force 100 N is 50 N at 30° as shown in figure, what is the magnitude and direction of the other force Q?



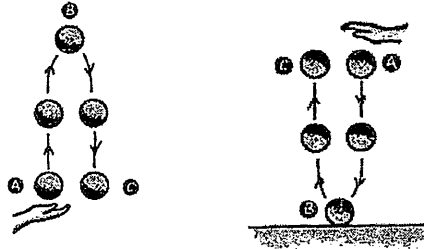
- (c) What do you mean by a perfect truss, deficient truss, and redundant truss? Identify the type of the truss given in figure.



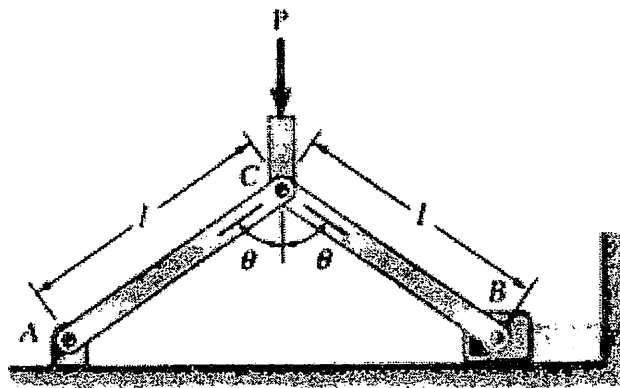
- (d) Find the volume of the torus shown in the figure which was formed by revolving the circular region bounded by $(x-2)^2 + y^2 = 1$.



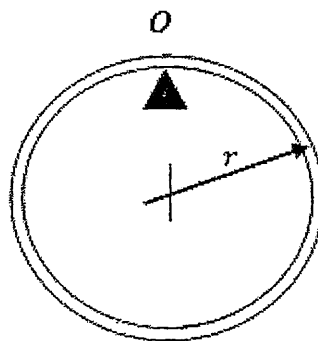
- (e) Plot the variation of acceleration, velocity, speed, and displacement with respect to time when a ball is (i) projected vertically up with a velocity U (ii) released from a height h on to cement floor as shown in figure. Take the release point of the ball as the reference.



- (f) Determine the force exerted by the vice on the wooden plank when a force $P = 200 \text{ N}$ is applied at C as given figure, using the principle of virtual work. Take the value of angle $\theta = 30^\circ$ and $l = 500 \text{ mm}$.



- (g) State and prove the work-energy principle in curvilinear translation.
- (h) A thin circular hoop of mass per unit length m and mean radius r hangs over a pin at O at its circumference as shown in figure. Find the period of small oscillations of the hoop in the vertical plane.

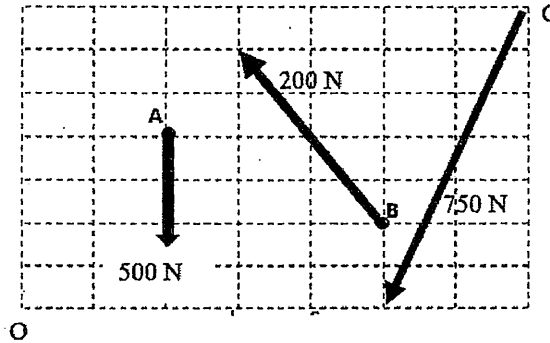


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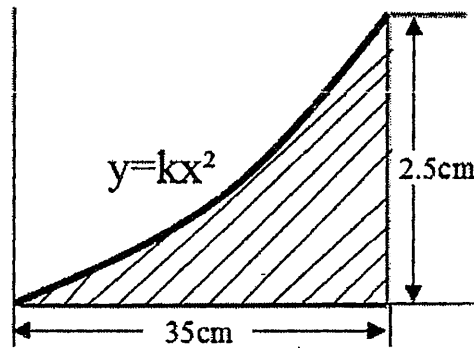
PART B

(4 × 12 = 48)

- II. (a) Determine the magnitude, direction, and line of action with respect to the origin O of the resultant of the forces acting at A, B, and C shown in figure. Take each grid as a square of size 1 cm × 1 cm. (6)

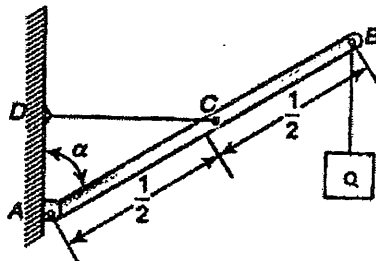


- (b) Locate the centroid of shaded area as shown in figure with respect to the x-y axis starting from the first principles. (6)

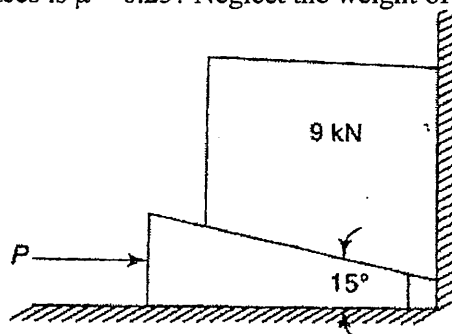


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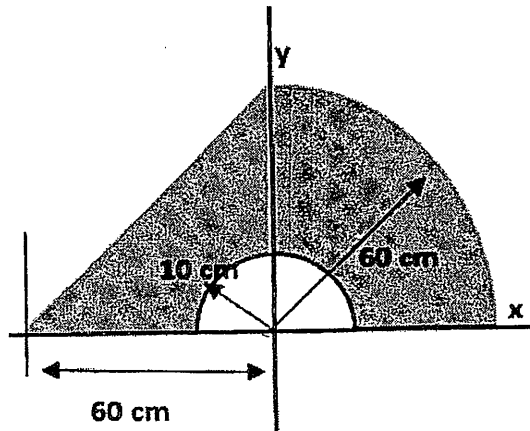
- III. (a) A rigid bar AB is supported in a vertical plane and carries a load $Q = 100 \text{ N}$ at its free end as shown in figure. Neglecting the weight of the bar itself, compute the magnitude of the tensile force S induced in the horizontal string CD and the support reactions at A if the length of the bar $l = 50 \text{ cm}$ and the angle $\alpha = 60^\circ$. (6)



- (b) To raise a heavy stone block weighing 9 kN, the arrangement shown in figure is used. What horizontal force P will it be necessary to apply to the wedge in order to just raise the block if the coefficient of friction for all contiguous surfaces is $\mu = 0.25$? Neglect the weight of the wedge. (6)

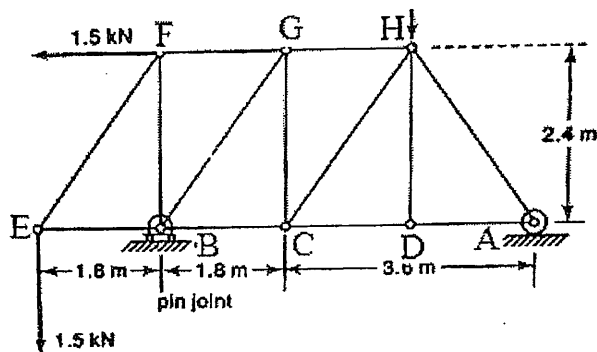


- IV. Calculate the moment of inertia of the shaded area shown in figure with respect to its centroidal x-y axes. (12)

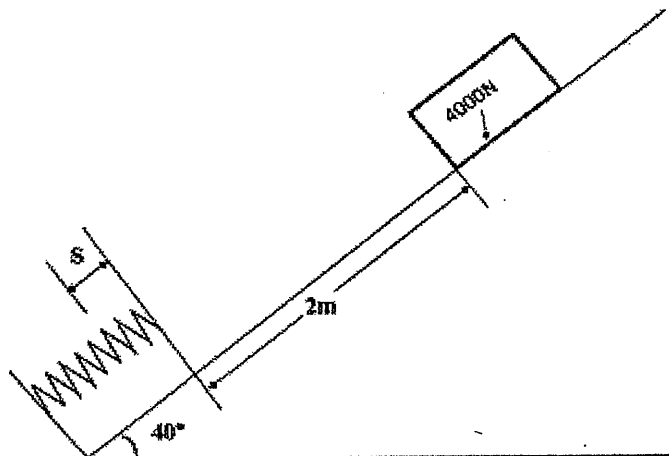


OR

- V. Determine the axial forces in the bars of the truss loaded and supported as shown in figure. (12)



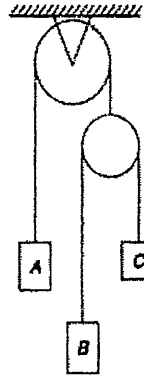
- VI. (a) A train starting from rest increases its speed from 0 to 100 km/hr with a constant acceleration 10 m/s^2 , runs at this speed for some time, and finally comes to rest with constant deceleration 8 m/s^2 . If the total distance travelled is 50 km, find the total time t elapsed. (6)
- (b) A block weighing 4000 N slides down a 40° incline as shown in figure. After moving 2 metre it strikes a spring whose stiffness is 20 kN/mm. If the coefficient of friction between the block and the incline is 0.2. Determine the maximum deformation of the spring. (6)



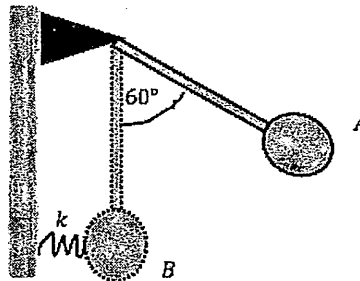
OR

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- VII. A system of weights and pulleys is arranged in a vertical plane as shown in figure 15. Neglecting friction and the inertia of the pulleys, find the acceleration of each weight if their magnitudes are in the ratio $W_a : W_b : W_c = 3 : 2 : 1$. (12)



- VIII. (a) Two adjacent missile launchers fire two missiles at 50 m/s simultaneously at angles of elevation α_1 and α_2 for the same target at a range of 1000 m. Calculate the time difference between the hits. (6)
- (b) A simple pendulum of weight $W = 5$ N and length $l = 25$ cm is released from rest at A, swings downward under the influence of gravity and strikes a spring of stiffness $k = 1200$ N/m at B. Neglecting the mass of the spring, determine the compression that it will suffer. (6)



OR

- IX. A homogeneous square plate of weight $W = 100$ N and side $a = 10$ cm hangs in a vertical plane by two pins A and B as shown. Calculate the horizontal and vertical components of the reaction at A an instant immediately after the pin at B is removed. (12)

